Evaluation Integrity in Multi-Agent Slice Admission Control: When Compliance Metrics Mask Training Collapse

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***Abstract*—Reinforcement-learning-based slice admission control is normally evaluated by whether each decision keeps its slice inside its Service Level Agreement (SLA) margin. We show this assumption can fail outright in a realistic contention regime, on a real Open RAN (O-RAN) ratio-ceiling admission-control surface extended from the live, single-gNB deployment validated in our prior work to a coordinated multi-gNB cluster: an episode-level SLA-compliance metric scored a degenerate, always-accept policy no worse than genuinely differentiated decision-making, because blocking a low-priority slice under congestion – the reward-optimal action – costs compliance without ever recovering it once a stress episode is underway. The metric could not tell a training collapse from correct behaviour. This evaluation-integrity failure mode, a correctness-aware metric pair (mean reward per step and block precision, both derived from data every arm already logs) that exposes it, and a reproduction-versus- replication discipline that catches a second, independent class of evaluation error, are this paper's central contributions. We build the evidence on GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture (a shared Graph Attention Network encoder over the gNB topology feeding per-slice Q-heads) that we use as a *substrate* for the evaluation study, evaluated against an independent per-gNB DQN ablation and a flattened-state single-agent DQN baseline over a 30-seed offline stress-regime campaign – the architecture is the vehicle through which the metric failure and its fix are demonstrated, not the paper's headline result. Our initial GAT encoder, lacking any normalisation, collapsed to a trivial "accept every request" policy on all 30 seeds, invisibly to the compliance metric; two encoder fixes – per-layer LayerNorm, then per-slice Q-heads that isolate each slice's gradient signal – reduce the collapse rate from 30/30 to 9/30 seeds, a residual-reliability gap we report rather than paper over. On the correctness-aware metric, GAT- CTDE decisively beats the independent-DQN ablation that isolates the same GAT/centralised-training contribution (+1.461, 95% CI [0.550, 2.782], Wilcoxon $p < 0.0001$, 27 wins / 0 ties / 3 losses), but its advantage over our existing single-agent baseline is small and not statistically decisive at this sample size (+0.195, 95% CI [$-$0.017, 0.442], Wilcoxon $p = 0.0577$, 20 wins / 3 ties / 7 losses out of 30 seeds) – a more modest architectural finding than we originally reported, corrected after catching a second data-integrity bug with the same discipline the collapse diagnosis required (Section V), and reported honestly as modest rather than inflated to fit a headline-architecture narrative. Extending to a federated variant with DP-SGD-style privacy, we find federating the architecture costs nothing measurable in reward at this sample size (+0.133, $p = 0.8125$), while differential privacy shows a real threshold effect on block precision – perfect targeting through noise multiplier $\sigma = 1.0$, dropping sharply by $\sigma = 4.0$ – rather than the compliance metric's misleading, non-monotonic curve at the same noise levels: a second, independent instance of the same evaluation-integrity failure this paper's central contribution targets. Evaluating the same frozen policies under mid-episode gNB dropout, demand spikes, and agent churn, dropout and churn both produce a real, accelerating cost with severity across every arm, while demand spikes do not threshold at all on block precision, a genuinely different failure mode. Finally, we cross-check every result in this paper against an independent seed sample, not only the same seeds reproduced again – the reproduction-versus-replication discipline this paper argues evaluation practice needs – and report one finding this caught: an initial reading that independent-per-gNB DQN is architecturally immune to agent churn did not survive replication and is retracted here rather than reported. Extending the cluster from 3 to 19 gNBs under ring and hex-grid adjacency, single- agent DQN's always-accept collapse becomes total (15/15 eval runs across two disjoint seed samples, confirmed at the individual-decision level), GAT-CTDE collapses only partially, and independent DQN never collapses but never cleanly differentiates either – a real, cluster-size-specific effect distinct from anything at $N = 3$. GAT-CTDE's own collapse rate initially disagreed sharply between two seed samples (31% vs. 78%); a third, independent sample drawn specifically to resolve the discrepancy converged on 46% [29%, 65%] across 21 seeds, showing both smaller samples were honest reads of a genuinely wide distribution rather than one being wrong. Finally, we anchor this paper's single-agent baseline against a real OAI testbed: its offline-trained differentiated-shedding decisions transfer to live traffic unchanged (replicated across two independent runs), while its SLA-margin metric diverges from anything offline by orders of magnitude – traced, not left unexplained, to a real backlog regime this rig's own prior live data independently corroborates, and narrowed from six orders of magnitude to three with a physically- grounded, latency-based recalibration. Together these results argue that admission-control evaluation in multi-agent, privacy-constrained, disruption-prone settings needs metrics tied to the actual training objective, not only to a downstream SLA proxy that can reward the wrong policy, and that architectural claims need more than one seed sample before they are trusted.**

***Index Terms*—5G, Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Differential Privacy, Open RAN**

I. Introduction

Network slicing lets a single physical radio access network serve heterogeneous service classes – Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) – under distinct per-slice Service Level Agreements (SLAs). Our earlier work [1], [2] established that a Deep Q-Network (DQN) admission controller can learn to enforce

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these SLAs under a single gNodeB (gNB), and validated this on a live OpenAirInterface (OAI) testbed under a Quality-of-Experience-aware reward.

A real multi-cell deployment does not admit requests at one gNB in isolation: neighbouring cells share backhaul, spectrum planning, and – in the reward formulation we build on here – a cluster-wide load-balance objective. Coordinating admission decisions across gNBs is therefore a genuinely multi-agent problem, and the natural architectural question is whether a shared, graph-structured representation of the cluster's state helps a set of per-gNB agents make better joint decisions than either a single agent using the flattened joint state, or independent per-gNB agents with no shared representation at all.

This paper's contribution is evaluation integrity, not architecture, and its four parts are ordered accordingly. First, we identify an evaluation-integrity failure mode specific to stress-regime admission control: the standard episode-level SLA-compliance metric can score a degenerate, always-accept policy no worse than genuinely differentiated decision-making, because blocking a low-priority slice under congestion – the reward-optimal action – costs compliance without ever recovering it once a stress episode is underway, so the metric cannot distinguish a training collapse from correct behaviour. Second, we introduce a correctness-aware metric pair (mean reward per step and block precision, both computed from data every training run already logs, requiring no new instrumentation) that exposes this gap directly, and use it to diagnose and fix the collapse we found: two encoder fixes reduce the collapse rate from 30/30 to 9/30 seeds, a residual reliability gap we report as a first-class finding rather than smoothing over. Third, we establish a reproduction-versus-replication discipline – checking every headline result against an independent seed sample, not only reproducing the same seeds – and show it catches a second, distinct class of evaluation error that same-seed reproduction cannot: a real architectural overclaim under mid-episode disruption (Section IX) that read as confirmed on one seed sample and was retracted once a disjoint sample contradicted it. Fourth, we build the empirical substrate all three of the above are demonstrated on: GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture (a Graph Attention Network [3] encoder over the gNB topology, shared across agents and trained centrally, feeding decentralised per-slice Q-heads), on the real O-RAN ratio-ceiling admission-control surface extended from the live single-gNB anchor validated in our prior work [1], [2], extended to a federated training regimen with differential privacy and evaluated under mid-episode gNB dropout, demand spikes, and agent churn, then to cluster sizes $N \in \{7, 19\}$ under ring and hex-grid adjacency, and finally anchored against a real, single-gNB OAI testbed. We label these five campaigns M2 (centralised), M3 (federated/privacy), M4 (disruption), M6 (cluster-size scaling), and M8 (live single-gNB anchor) throughout, matching this project's internal milestone numbering (M5 is this paper's own evaluation-integrity reframe, not a separate results campaign; M7's two results – a FedProx-under- heterogeneity null finding and a collapse-rate-narrowing follow-up – are summarised in Section IV and folded into Section X respectively, rather than given a section of their own). A sixth, earlier milestone (M1, live-testbed recalibration) motivates the offline stress environment of Section III and the bounded scope of Section XI, but reports no results of its own here. The architecture is deliberately not this paper's headline: it is the vehicle that makes the metric failure, its fix, and the reproduction-replication gap concrete and measurable, and Section II positions it against the closest prior art to make that scoping explicit before any results are reported.

II. Literature Review

Our own systematic review [4] screened 6,286 records down to 67 core 2024–2026 studies on machine-learning-driven, user-centric distributed network slicing, organised around three enabling pillars – topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation – and found individual pillars well developed but rarely combined under realistic constraints, and never with a validated live deployment. That review supplies this section's broader landscape; here we go narrower and ask a different question of the closest works in each pillar, plus several specifically-positioned pieces outside that review's 2024–2026 window: does any of it examine whether its own reported evaluation metric can mask a degenerate policy, or check a headline finding against an independent seed sample? We find no work that does either, which is the gap this paper fills.

A. Graph-Attention State Representation for Slicing

The canonical result behind our own encoder choice is Shao *et al.* [5], who formulate dense-cellular slicing resource management as MARL with one agent per base station and show a graph attention encoder strengthens inter-agent cooperation over plain DQN/A2C baselines – the architectural precedent GAT-CTDE's encoder builds on directly. Closest to our own admission-control setting specifically, Ahmadi *et al.* [6] extend a coordinated multi-agent slicing-and-admission line of work (discussed further below) to generalise across unseen RAN/MEC topologies via graph attention, without retraining – a topology- generalisation question distinct from, but complementary to, the evaluation-integrity question this paper asks on a single fixed topology. We test the complementary direction directly (Section X): whether GAT-CTDE's own correctness-aware metrics remain reliable as cluster size and adjacency sparsity change, rather than whether the policy generalises to an unseen topology without retraining. Beyond admission control specifically, graph learning has been applied to joint user association and resource allocation [7], intent-driven RAN optimisation [8], transferable bandwidth allocation via meta-learning [9], latency-optimal digital-twin placement [10], general adaptive network optimisation [11], offline-data delay-sensitive resource allocation [12], QoS prediction [13], and non-stationary space-air-ground-sea slicing [14]. None of these report a metric that structurally rewards a degenerate policy, because none instrument the correctness-aware pair this paper introduces to look for one.

B. Multi-Agent RL for Coordinated Slicing and Admission Control

Sulaiman *et al.* [15] is the closest prior art to our own problem: multi-agent DRL jointly solving slicing and admission control, coordinated across the substrate network, from the same research group as the topology-generalisation follow-up above. Neither paper in that line of work reports whether their own reward-optimal policy could be mistaken for a degenerate one under their own evaluation metric, nor checks a headline result against an independent seed sample – both are exactly what this paper adds on a comparable problem. Related MARL work spans sample-efficient inter- slice orchestration [16], constrained MARL for flexible O-RAN slicing [17], constrained traffic-aware MARL for joint URLLC admission and resource provisioning directly in O-RAN [18], [19] (the closest single-column ablation to our own independent-per-gNB-DQN arm), sharpness-aware robustness [20], energy- efficient decentralised cooperation [21], MATD3 with evolutionary-game user association [22], retraining- free scalable slicing [23], and reliability- enhanced VNF orchestration [24]. Two very recent works bear directly on our M4 disruption campaign (Section IX): Tashman and Cherkaoui [25] study adversarial jamming attacks and SLA-violation recovery in AI-driven RAN slicing, and Fatehi *et al.* [26] apply interpretable attention-based multi-agent PPO to resolve latency spikes in 6G RAN slicing. Both study robustness to a disruptive condition, as we do; neither checks whether the metric reporting that robustness is itself trustworthy under a stress regime, and neither replicates a headline robustness finding on an independent seed sample – the exact gap our M4 campaign's independent-seed replication (Section IX-D) was built to close, and the exact gap that replication shows is not hypothetical: it is what overturned our own initial churn-immunity reading.

C. Federated and Privacy-Preserving Learning for Slicing

Zhang *et al.* [27] coordinate two independent O-RAN xApps (power control and slice resource allocation) via federated DRL – the closest prior art to our own M3 federated variant's motivation (periodic, not per-step, cross-node synchronisation for disaggregated RIC deployments), though without a differential-privacy step or a correctness-aware metric to check whether federating changed decision quality, not just a throughput/ delay average. Yasin *et al.* [28] combine federated edge learning with differential privacy specifically to secure the O-RAN RIC – the closest FL+DP precedent to our own – but target a different threat model (securing the controller against compromise) rather than characterising the privacy-utility trade-off of a control policy itself, which is what our σ-sweep (Section VIII) reports. Further federated work covers VNF splitting [29], prediction-based slice mobility [30], personalised per-slice privacy [31], cooperative-game network selection [32], elastic multi-task FL over O-RAN [33], DQN-enhanced FL resource allocation [34], safe energy-efficient FL orchestration [35], [36], and FL-assisted freshness optimisation [37]. None of these report a privacy-utility curve on a metric verified, as ours is, to survive the same collapse-masking failure mode Section V diagnoses.

D. Positioning

A small set of works begins to combine two of the three pillars – graph-attentive MARL for SLA-aware orchestration [38], GNN-based multi-agent DRL for multi-domain orchestration [39], Transformer-GNN federated prediction [40], and federated multi-agent DDPG for MEC slicing [41] – but, like every other work surveyed above, evaluate exclusively on their own reported outcome metric, computed once, on one seed or one run. Of the works reviewed here and in [4], none report checking whether their evaluation metric could reward a degenerate policy, and none report an independent-seed replication of a headline finding. This is the gap this paper fills: not a new point in the GAT/MARL/FL design space, but a demonstration – on GAT-CTDE, extended to federation, privacy, and disruption, as the substrate – that both checks are necessary, and that skipping either one changes what a reader is entitled to believe the resulting numbers mean.

III. System Model and Problem Formulation

A. Cluster Topology and Admission Decisions

We consider a cluster of $N{=}3$ gNBs, each serving the same three slices (URLLC, eMBB, mMTC) under a shared per-gNB Physical Resource Block (PRB) budget B. As in our single-gNB work, an admission decision is realised as a PRB-ceiling adjustment (`slicing_control_m` min/max ratio), not a literal per-flow accept/reject at the scheduler, since the real OAI xApp control API only exposes ratio ceilings. No real multi-gNB physical topology (inter-site distances, backhaul graph, neighbour relations) is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected – every gNB attends to every other gNB's state – and note this explicitly as a modelling choice rather than a measured fact.

At each control step, every gNB has zero or more pending admission requests, each tagged with a slice identity. An agent decides accept/reject for each pending request independently, conditioned on the current cluster state; the discrete action space matches our single-gNB work exactly (`action_dim=2`), so any coordination benefit found here is attributable to the multi-agent architecture, not to a richer action space.

B. Reward and the Differentiated-Shedding Regime

The reward combines a per-slice SLA-compliance term with a cluster-wide load-balance penalty, weighted by a per-slice `priority_weight` (URLLC highest, mMTC lowest) and a congestion coefficient that was tuned, in prior work, so that the reward-optimal policy under congestion is *differentiated shedding*: continue accepting URLLC, mostly accept eMBB, and reject mMTC. This is not an assumption we make for this paper – it is a property of the existing reward weights, confirmed directly by Q-value inspection on a working baseline policy (Section V), and it is the behaviour against which we judge whether a new architecture is doing what it is supposed to do, not merely whether it scores well on a downstream proxy.

C. Offline Stress-Regime Environment

Every result in this paper except Section XI's bounded live anchor is obtained in an offline, closed-loop simulation environment deliberately oversubscribed to roughly 110% of one gNB's PRB budget across its three slices, reusing our existing multi-gNB configuration unmodified. A separate investigation (held out of this paper's scope) found that this offline environment's demand process does not reproduce the live testbed's SLA-margin distribution closely enough to support a live-rank prediction claim, even after recalibrating the demand model's temporal structure. We therefore use it here explicitly as a *stress environment* for the contention regime our live rig does not reach, not as a claim that any ranking found offline would reproduce live – Section XI tests this paper's own single-agent baseline against a real gNB directly rather than relying only on prior work's anchor, and finds exactly this limit again, more starkly, on a metric this paper computes itself. Multi-gNB claims in this paper remain offline-only throughout; no rig with more than one physical gNB exists to test them against.

IV. Methodology

A. GAT-CTDE: Shared Encoder, Decentralised Q-Heads

Fig. 1(a) shows the architecture. At every control step, each gNB's own per-slice [PRB-used-ratio, congestion-level, queue-length-norm] features form that node's row in a joint node-feature matrix, mirroring exactly the per-(gNB, slice) triple our existing multi-gNB state encoding already computes – kept as a per-node matrix here instead of flattened, since the GAT encoder needs graph structure, not a flat vector. A two-layer, four-head GAT [3], implemented directly (the small graph sizes here make a dense, masked-attention implementation as fast as a sparse one and avoid an external dependency), produces one embedding per gNB. Centralisation comes from this encoder: its parameters are updated by every agent's loss jointly, so it sees the full cluster's state and structure during training. Decentralised execution comes from what happens next: at action-selection time, each agent consumes only its own node's embedding row to decide its own pending requests – never another agent's row.

1) Per-Slice Q-Heads: A first version of this architecture fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head MLP. Under this design, mmTC's true reward-optimal margin under congestion is small by the reward's own calibration (a net expected value on the order of -1 against Q-values in the hundreds, a consequence of this architecture's per-request temporal-difference scheme), while URLLC's much larger priority weight produces proportionally larger gradients from URLLC-context samples. Because all three slices' samples update the *same* shared parameters, URLLC's abundant, large-magnitude gradients can overwrite whatever fine adjustment mmTC's much smaller true signal needs before it registers – a standard shared-parameter/class-imbalance failure mode. We replace the single shared head with *one small MLP per slice*, selected by the context one-hot at the point where the head is applied (Fig. 1(a)): each slice's decision boundary now has its own parameters, so one slice's gradients can no longer directly overwrite another's. The shared GAT encoder upstream is unaffected and remains the architecture's centralising element.

Algorithm 1 GAT-CTDE: centralised training, decentralised execution

Require: shared GAT encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}_{k=1}^{K}$, target networks $f_{\theta^-}, \{Q_{\phi_k^-}\}$, replay buffer $\mathcal{D}$, fully-connected adjacency A, discount γ, target-sync period C episodes

1: **for** episode $= 1, 2, \ldots$ **do**
2: reset environment; observe joint node features $X_1 \in \mathbb{R}^{N \times d}$
3: **for** step $t = 1, \ldots, T$ **do**
4: $E_t \leftarrow f_\theta(X_t, A)$ {one embedding row per gNB; centralised: sees every node}
5: **for** each pending request r at gNB i, slice k **do**
6: $a_r \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$ w.p. $1-\epsilon$, else uniform random {decentralised: own row $E_t[i]$ only}
7: **end for**
8: $X_{t+1}, \rho_t, \text{done} \leftarrow \text{ENV.STEP}(\{a_r\})$
9: store $\big(X_t, X_{t+1}, \{(i, k, a_r)\}_r, \rho_t, \text{done}\big)$ in $\mathcal{D}$
10: **if** $|\mathcal{D}| \geq$ warmup size **then**
11: sample minibatch $B \subset \mathcal{D}$
12: $E_B \leftarrow f_\theta(X_B, A)$; $E'_B \leftarrow f_{\theta^-}(X'_B, A)$ {target net, no grad}
13: **for** each request $(i, k, a) \in B$ **do**
14: $q \leftarrow Q_{\phi_k}(E_B[i])[a]$
15: $y \leftarrow \rho + \gamma \cdot (1-\text{done}) \cdot \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
16: **end for**
17: $\mathcal{L} \leftarrow$ mean Huber loss over every request's (q, y) pair {per-request, never summed per step}
18: update θ, $\{\phi_k\}$ by $\nabla\mathcal{L}$; clip grad-norm to 1.0
19: **end if**
20: **end for**
21: **if** episode $\bmod C = 0$ **then**
22: $\theta^- \leftarrow \theta$; $\phi_k^- \leftarrow \phi_k \ \forall k$
23: **end if**
24: **end for**

2) Training Procedure: Algorithm 1 states the training loop precisely: the encoder is updated from every request in a sampled minibatch, regardless of which gNB or slice it belongs to (centralised training), while action selection at rollout time consults only the acting agent's own embedding row (decentralised execution). The per-request TD formulation on line 13 is the fix described above in the context of the architecture's earlier, unstable QMIX-style variant (Section V's companion bug, documented in full in `ctde_policy.py`): every pending request in a step is its own independent temporal-difference sample, never summed or averaged across a step before the loss is computed, matching our existing single-agent DQN's own granularity exactly.

B. Federated Training with Differential Privacy

Fig. 1(b) shows the federated variant. Each gNB becomes an FL client with a private copy of the same GAT-CTDE network (minus the unused QMIX-style mixer our centralised

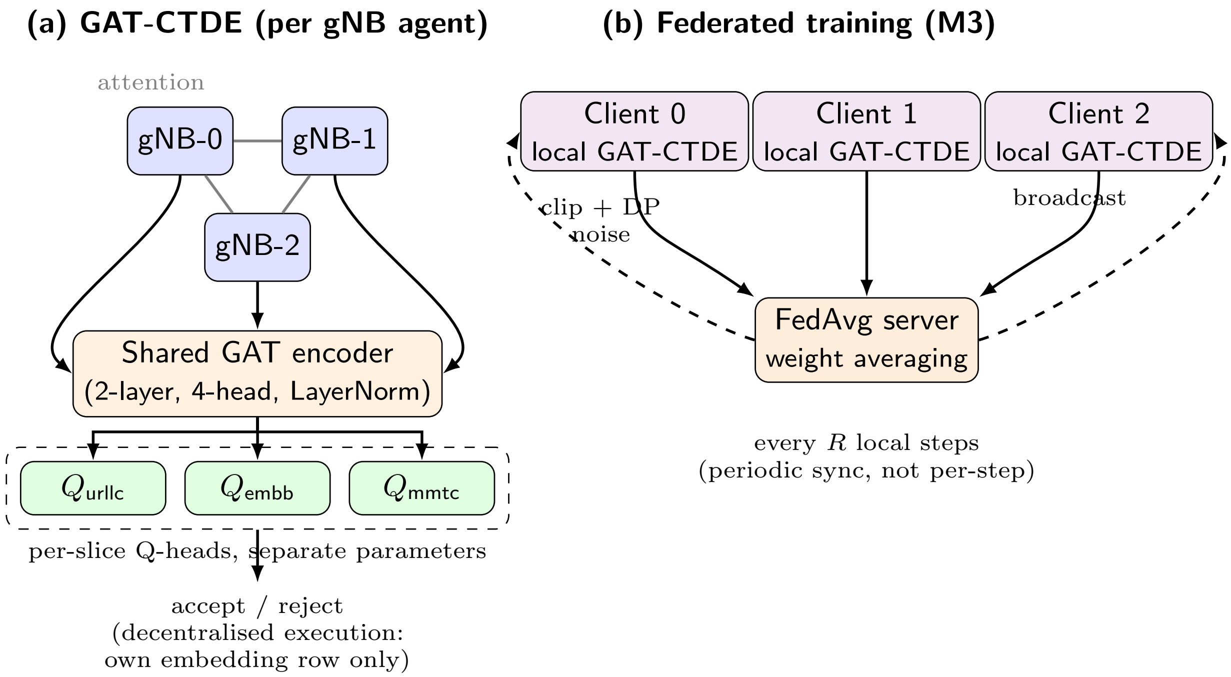

Fig. 1. GAT-CTDE architecture. (a) A shared Graph Attention encoder over the fully-connected 3-gNB topology produces one embedding per node; per-slice Q-heads, each with its own parameters, consume that node's own embedding row at action-selection time (decentralised execution). (b) The federated training variant: each gNB is a client with a private copy of the same network, synchronised via periodic FedAvg with an optional per-client DP-SGD clip-and-noise step.

training does not require), initialised identically by a server broadcast. Every client trains only on its own requests – transitions are partitioned by owning agent, so no client ever sees another client's raw admission data – while its GAT encoder still consumes the full joint node-feature snapshot, the same public per-slice occupancy information every other arm in this paper observes. Every R local training steps, the server averages all clients' online weights (FedAvg [42]; weighted equally, since every client runs the same amount of local training) and broadcasts the result back. We also implement FedProx [43], which differs only in the client-side objective (a proximal term pulling local weights toward the last broadcast global model). At the time of the M2/M3 campaigns we believed our topology treated every gNB as an interchangeable peer with no client heterogeneity for FedProx's correction to address, so a full sweep would have evaluated a fix for a problem this environment did not have. Section X corrects this belief – every client already draws offered demand from an implicit, seeded per-gNB load multiplier neither this section nor the M2/M3 campaigns had deliberately controlled – which motivated a FedProx sweep under both a genuine homogeneous-load control and this already-present heterogeneity ($\mu \in \{0.01, 0.1, 1.0\}$, two orders of magnitude): it found no measurable dividend at any tested strength, traced to a concrete mechanism rather than left unexplained (real per-round client drift within `local_steps_per_round` averages 0.03% of the TD loss, too small for the proximal term to matter regardless of load heterogeneity) – summarised in Section XIII, full investigation in the project's supplementary milestone documentation.

Differential privacy is applied as a standard DP-SGD [44] step: every client's gradient is clipped to a fixed norm and, when enabled, perturbed with calibrated Gaussian noise scaled by a noise multiplier σ, immediately before that client's own optimiser step – so the local update itself is privatised, not only the round-boundary aggregate. We report σ as the primary privacy parameter throughout. A formal (ε, δ) bound is also reported, computed via zero-concentrated differential privacy (zCDP) [45] composition of the Gaussian mechanism: with $\rho = k/(2\sigma^2)$ over k per-client releases, converted via $\varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}$. This is a conservative, non-subsampled composition bound, not a tight moments-accountant figure – we report it alongside σ, not in place of it.

Algorithm 2 states one federated round precisely. Every client runs the identical per-request TD loss Algorithm 1 already defines (line 8's FedProx term is additive and zero whenever μ=0, our reported configuration); clipping is applied identically whether or not noise is added (line 11 vs. line 9), so a σ=0 run and a σ>0 run differ only in whether noise is injected, not in any other aspect of the client's update – the isolation the privacy- cost comparison in Section VIII depends on.

V. A Training Collapse, Diagnosed and Fixed

A. Discovery

While generating a privacy-utility sweep for the federated variant (Section VIII), noise levels that should have produced genuinely different, stochastically-trained checkpoints instead produced bit-identical per-seed evaluation compliance. Tracing

Algorithm 2 Federated GAT-CTDE round with DP-SGD

Require: N clients, each with local network $(\theta_i, \{\phi_{i,k}\})$; global snapshot $\bar{\theta}$; local steps per round R; clip norm C_{dp}; noise multiplier σ; FedProx weight μ $(= 0$ unless noted$)$

for round $= 1, 2, \ldots$ **do**
  for local step $= 1, \ldots, R$, each client i independently **do**
    sample minibatch B_i of i's own requests from i's local buffer
    $E \leftarrow f_{\theta_i}(X_{B_i}, A)$ {full joint state; client's OWN local weights only}
    $\mathcal{L}_i \leftarrow$ per-request Huber TD loss (Alg. 1, lines 11–13) $+ \frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$ {FedProx term}
    $g_i \leftarrow \nabla_{\theta_i, \phi_i} \mathcal{L}_i$
    if $\sigma > 0$ **then**
      $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$; $g_i \leftarrow g_i + \mathcal{N}(0, (\sigma C_{\text{dp}})^2 I)$ {DP-SGD [44]}
    else
      $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$ {same clip norm, no noise – isolates the privacy cost}
    end if
    $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta\, g_i$
  end for
  $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N}\sum_{i=1}^{N}(\theta_i, \phi_i)$ {FedAvg [42], equal client weights}
  for each client $i = 1, \ldots, N$ **do**
    $(\theta_i, \phi_i) \leftarrow (\bar{\theta}, \bar{\phi})$ {broadcast}
  end for
end for

this back, we found that our first GAT-CTDE encoder's eval policy blocked zero requests across every evaluation episode, on all 30 campaign seeds – an always-accept policy is a functional no-op, so any two checkpoints that reach it produce identical environment trajectories regardless of their internal weights.

B. This Was Not the Reward-Optimal Policy

We confirmed, rather than assumed, that always-accept was a genuine failure and not a legitimate optimum: a direct Q-value probe on the collapsed checkpoint, under a synthetic congested state, showed $Q(\text{accept}) - Q(\text{reject})$ positive and large for every slice, including mmTC – the same broken signature our reward-tuning process had already documented and fixed once before, for a different architecture, under different (already-superseded) reward weights. Under the current, correctly-calibrated weights, the independent-DQN ablation (Section VII), which sees identical raw features through no encoder at all, does not exhibit this collapse – isolating the encoder, not the reward or the training loop, as the fault.

C. Root Cause: Magnitude, Not Pattern

Comparing the encoder's own embedding for a synthetic idle state against a congested one on the collapsed checkpoint: the raw input difference has ℓ_2 norm 2.55, and the resulting embedding difference has norm 19.6 – an apparently large, congestion-sensitive response. But the cosine similarity between the two embeddings is 0.99997: the encoder was encoding congestion almost entirely as embedding *magnitude*, with no mechanism anywhere in the GAT layers preventing this, and no rotation of direction to speak of. Since the downstream Q-head is a roughly linear function of its input, larger magnitude pushes $Q(\text{accept})$ up faster than $Q(\text{reject})$ as congestion increases – backwards from the reward's intent.

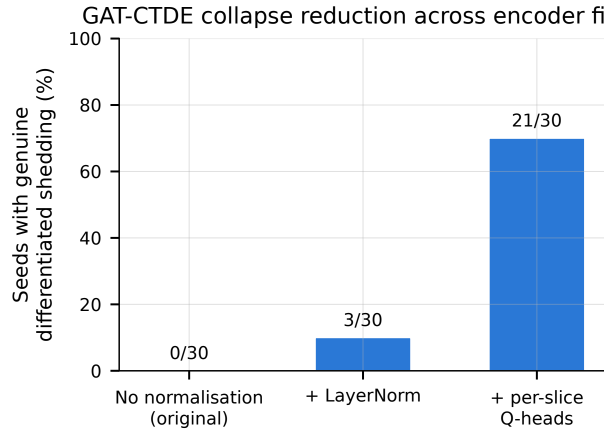

Fig. 2. Fraction of the 30 campaign seeds reaching genuine, correctly-targeted differentiated shedding (mMTC-only blocking under congestion), across the original unnormalised encoder and the two fixes described in Section V.

D. Fixes

We add a per-layer `LayerNorm` to the GAT encoder, constraining every node's embedding to zero-mean, unit-variance regardless of input magnitude. This is a real, verified improvement – collapse drops from 30/30 to 27/30 seeds – but not a complete fix. We then apply the per-slice Q-heads described in Section IV-A1, which reduces collapse further, to 9/30 seeds (Fig. 2). We verified each fix on a small pilot before committing to a full campaign's compute budget, and we tested one further variant, disabling LayerNorm's learnable affine transform, that made results *worse* (0/3 pilot seeds differentiated, versus 1/3 with the affine transform enabled) and was reverted – not every plausible fix helped, and we report the one that did not.

E. A Second Bug: Eval-Log Contamination

A separate, unrelated bug was caught while building an evaluation-time disruption harness for planned future work (Section XIII): our logging utility opens its output file in append mode and never truncates it, and GAT-CTDE was retrained in place three times over the course of diagnosing and fixing the collapse above (the original encoder, the LayerNorm-only fix, and the per-slice-heads fix), each time writing evaluation results to the same per-seed path. Because nothing cleared that path between runs, every one of the 30 seeds' evaluation logs had silently accumulated all three runs' episodes stacked together – confirmed directly by each log's own episode index resetting to 1 twice. Every GAT-CTDE number reported through an earlier draft of this paper, and every M3 result that reuses GAT-CTDE's centralised evaluation as a reference point (Section VIII), was therefore a hidden three-way average across two architecturally-obsolete runs and the true final one, not the final architecture's behaviour alone. The saved model checkpoints themselves were unaffected (a different save routine overwrites rather than appends, verified by reproducing one seed's final-block statistics exactly from its checkpoint in isolation), so the fix required no retraining: we cleared every stale log and re-ran evaluation once, cleanly, against the already-trained checkpoints, and hardened the training script to clear its output path before any future run. All numbers in this paper reflect that corrected data; the magnitude of the correction is largest on exactly the paired comparisons this section's own diagnostic discipline exists to protect, which is why we report it here rather than silently revising the numbers.

F. Correctness-Aware Metrics

The fixes above make the architecture more correct while making its `sla_compliance_all_slices` score *worse* (Section VII): blocking mmTC under congestion, the reward-optimal action, does not recover mmTC's own SLA margin in this stress regime, so every newly-correct decision costs compliance the metric never credits back. We therefore report two further metrics, both computed from data every arm already logs, requiring no new instrumentation:

- **Mean reward per step**: the actual training objective, matching our existing evaluation harness's own established definition (per-episode mean of every step's logged reward, then mean across episodes) – since the reward's own calibration is what defines correct behaviour in the first place, this tests whether a policy does what it was trained to maximise, directly.
- **Block precision**: of every request an arm blocks, the fraction that target mmTC specifically, the only slice the reward calibration ever makes reject-optimal. Undefined, and reported as such, for seeds that never block anything.

VI. Experimental Setup

Both the centralised (Section VII) and federated (Section VIII) campaigns use 300 training and 50 evaluation episodes per seed, matching our existing single-agent training budget so that no comparison in this paper trades training time for a favourable result. The centralised campaign uses a fixed 30-seed list (seeds 900–929) with three arms: GAT-CTDE; an independent-DQN ablation (one per-gNB DQN instance, no parameter sharing, no topology information, the identical per-agent input GAT-CTDE's Q-head consumes) that isolates the GAT/centralised-training contribution specifically; and a single-agent DQN baseline, our existing architecture run unmodified against the flattened joint state, our today's established multi-gNB pattern. All three arms use matched hyperparameters (discount factor $\gamma = 0.95$, per-episode epsilon decay), a parity fix applied after an earlier pass found the two new architectures had inadvertently inherited a raw-DQN default schedule instead of our tuned one. The federated campaign uses the first 10 of these 30 seeds, the same per-seed training budget, and a five-level noise-multiplier sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its σ=0 level is directly comparable to the centralised campaign's same 10 seeds without re-running that arm. All confidence intervals are 95% bootstrap percentile intervals (10,000 resamples, not a normal approximation, since compliance is bounded on $[0, 1]$ and empirically right-skewed); paired comparisons use the Wilcoxon signed-rank test, matching this project's established preference for nonparametric methods.

TABLE I
M2 Campaign Results, 30 Seeds/Arm, 95% Bootstrap CIs.

Arm	Compliance	Mean reward/step	Block precision
GAT-CTDE	0.151 [0.083, 0.225]	**14.062** [13.91, 14.23]	0.952
Indep. DQN	0.049 [0.013, 0.093]	12.601 [11.30, 13.49]	0.901
Single-agent	0.115 [0.061, 0.176]	13.867 [13.69, 14.05]	0.987

VII. Results: Centralised GAT-CTDE Campaign

Table I and Fig. 3 report the final, post-fix 30-seed campaign. On `sla_compliance_all_slices`, GAT-CTDE's mean is 0.151, below its own pre-fix number (0.353) for the reason given in Section V-F – the metric penalises newly-correct mmTC-blocking decisions – and the paired comparison against single-agent DQN shows no clear direction: +0.036 (95% CI [−0.050, 0.126], Wilcoxon $p = 0.4549$, 11 wins / 9 ties / 10 losses). On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats the independent-DQN ablation decisively – +1.461 (95% CI [0.550, 2.782], $p < 0.0001$, 27/0/3), isolating a real contribution from the shared GAT encoder and centralised training – but its edge over single-agent DQN is small and does not clear conventional significance at this sample size: +0.195 (95% CI [−0.017, 0.442], $p = 0.0577$, 20/3/7). Its block precision (0.952) is high regardless, meaning that when GAT-CTDE does block a request, it is almost always the reward-correct slice. 21 of the 30 seeds now show genuine, correctly-targeted differentiated shedding (20 of those 21 blocking mmTC exclusively), up from 3/30 under the LayerNorm-only fix and 0/30 originally – a real, substantial behavioural improvement that, at this sample size, does not by itself translate into a decisive reward edge over the single-agent baseline specifically.

VIII. Results: Federated Training and Differential Privacy

A. Federation Cost

Comparing the centralised GAT-CTDE campaign against the federated variant with no DP noise, on the same 10 seeds: mean reward per step moves from 14.073 to 13.940, a paired difference of +0.133 (95% CI [−0.138, 0.470], Wilcoxon $p = 0.8125$) – no measurable cost to federating at this sample size (5 of the 10 seeds tie exactly, and the remaining seeds split 2 centralised / 3 federated). Both arms' block precision is perfect (1.000) on the seeds that block anything at all.

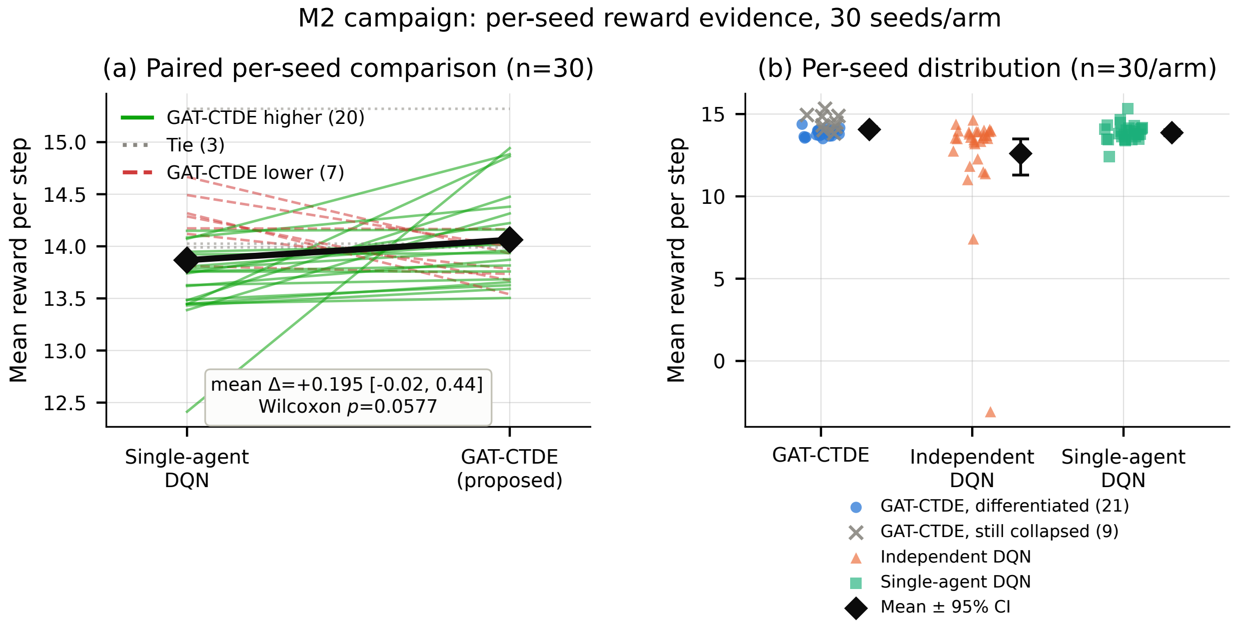

Fig. 3. 30-seed campaign, per-seed evidence for mean reward per step, the correctness-aware metric (Section V-F). (a) Paired per-seed comparison against single-agent DQN: each line is one seed, coloured by whether GAT-CTDE scored higher (green, 20), tied (grey, 3), or lower (red, 7) than the baseline on that seed – the same paired unit the Wilcoxon test is computed over, and the source of this figure's borderline $p = 0.0577$. (b) Full per-seed distribution for all three arms; GAT-CTDE is split into seeds that learned genuine differentiated shedding (blue, 21) versus seeds still stuck at the always-accept collapse (grey $\times$, 9) – shown without claiming the split predicts reward, since several still-collapsed seeds score among the arm's highest (always accepting never pays the reject cost).

B. *Privacy Cost: A Threshold, Not a Smooth Curve*

Fig. 4(b) reports block precision across the noise sweep. Precision is perfect through $\sigma = 1.0$, then drops sharply – 0.834 at $\sigma = 2.0$, 0.584 at $\sigma = 4.0$: differential privacy does not degrade decision quality gradually here, it holds until a real threshold and then degrades quickly. Mean reward per step, by contrast, stays in a comparatively narrow band across the whole sweep (12.45–14.19, with $\sigma = 2.0$'s own estimate also carrying the widest confidence interval of the five levels, [10.40, 13.99]), meaning DP noise does not crater the policy outright even at the highest tested σ – it erodes the *precision* of which slice gets blocked, not whether the policy tries to differentiate at all. The compliance-based sweep over the same data (not shown) is non-monotonic and its sign flips between adjacent σ levels; we do not report it as the primary privacy-utility curve for that reason, and note this as a second instance, independent of Section V's architectural finding, of `sla_compliance_all_slices` failing to track the quantity we actually care about.

IX. Results: Disruption Resilience

A. *Setup*

We evaluate every arm's already-trained, frozen checkpoint (no retraining) against three mid-episode disruptions, injected purely at evaluation time: *gNB dropout* (one randomly-chosen gNB's own requests are forced to reject and its row in the joint feature matrix every agent's encoder reads is zeroed, for a window); *demand spike* (a transient multiplier on the arrival rate, for a fixed 30%-of-episode window); and *agent churn* (one gNB's decisions are made by a freshly-initialised, never-trained copy of its own policy class instead of the loaded checkpoint, for a window). We call the third one *agent* churn rather than *client* churn deliberately: federation only happens during training, so a frozen, post-aggregation federated checkpoint has no per-client staleness left to reintroduce, and no arm shares cross-agent information at *inference* time beyond what each node's own encoder already reads from the joint state – there is no FL-specific desynchronisation left to disrupt. Replacing one agent's trained decision-making with an inexperienced stand-in is a different, well-motivated failure mode (active-but-uncoordinated vs. absent) that applies to every genuinely multi-agent arm; single-agent DQN, with no separable per-gNB agent, is excluded from this condition. Severity is the disruption window's duration (10%/30%/60% of the episode) for dropout and churn, and the arrival-rate multiplier ($2\times/4\times/8\times$) at a fixed window for spike. Target gNB and window start step are drawn fresh from each episode's own seeded random-number generator (RNG), not fixed to one gNB.

Because a demand spike changes request *volume*, raw mean reward per step is not a fair comparison against an undisrupted baseline – the reward's own service term is linear in how many requests were accepted that step (Section IV), so more requests mechanically inflates it regardless of decision quality. We therefore normalise by each step's own already-logged pending-request count before averaging (mean reward per pending request), built from data every run already logs, and use this normalised metric as the primary comparison throughout this section. Fig. 6 shows exactly where each

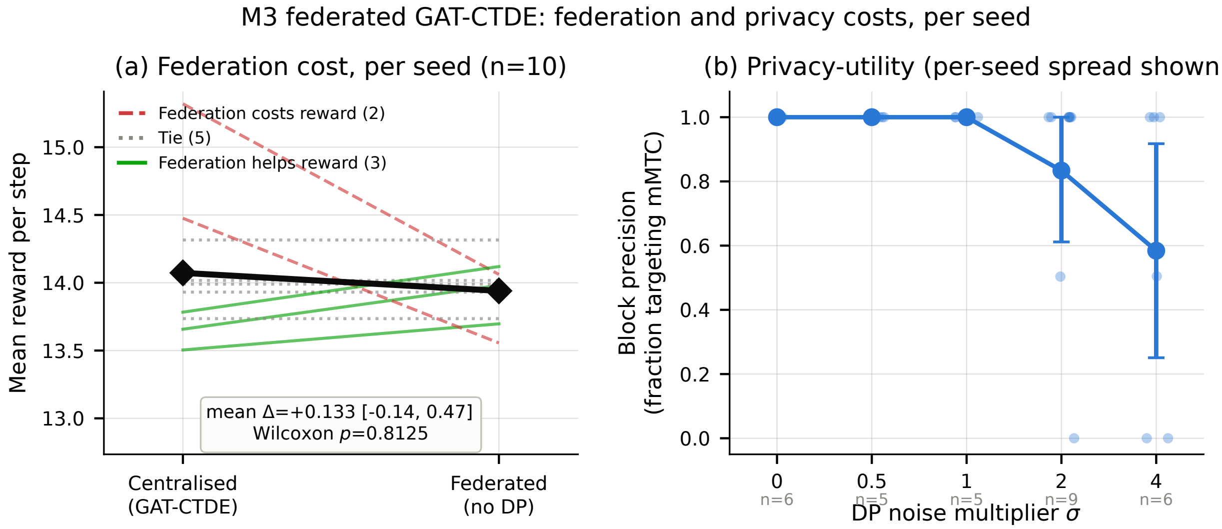

Fig. 4. Federated GAT-CTDE, per-seed evidence. (a) Paired per-seed federation cost on mean reward per step (same 10 seeds): centralised training scores higher on 2 seeds, federated on 3, with 5 ties – no measurable difference (Wilcoxon $p = 0.8125$), shown as individual seed trajectories rather than two independent bars, since the significance test itself is paired. (b) Privacy-utility curve using block precision, the metric that survives the compliance-based curve's non-monotonic, uninterpretable pattern at the same noise levels: individual seed values (light dots) are shown behind the mean, along with the seed count that ever blocked anything at each σ (precision is undefined otherwise) – precision is perfect through $\sigma = 1.0$ and drops sharply by $\sigma = 4.0$.

disruption kind intercepts the per-step control loop around the frozen policy.

B. Dropout and Agent Churn: A Real, Accelerating Threshold Effect

Fig. 5(a)-(b) report the normalised-reward cost at each severity. Dropout costs every arm significantly at every severity (Wilcoxon $p \leq 0.03$, 9 or 10 of 10 seeds hurt in every case), and the cost accelerates with severity rather than growing proportionally: for GAT-CTDE, +0.165 at the 10% window, +0.601 at 30% (roughly 3.6× the first increment), +1.710 at 60% (roughly 1.8× the second increment) – the same super-linear pattern holds for every arm. Agent churn shows the same accelerating shape for GAT-CTDE (+0.059/+0.200/+0.529, $p = 0.0039$ throughout) and the federated arm (+0.073/+0.240/+0.585, $p = 0.0039$ throughout).

Independent DQN's own churn cost needs both samples reported together to state honestly. On the seeds used throughout the rest of this paper (900–909), it is small and does not clear conventional significance (+0.000/+0.168/+0.640, $p = 0.084/0.106/0.065$) – which an earlier pass of this analysis read as independent DQN being architecturally immune to churn, reasoning that a policy which never attends to another agent's features (confirmed by inspection: it reads only its own node's row) has no coordination benefit a churned peer can take away. An independent-seed replication (seeds 1000–1009, disjoint from every other result in this paper) contradicts that reading directly: independent DQN's churn cost there is significant at every severity (+0.097/+0.657/+1.223, $p = 0.0020$ throughout, 10/10 seeds hurt each time) and *larger* than GAT-CTDE's or the federated arm's at matched severities in that sample – the opposite of immune. The direction was consistent across both samples throughout (churn always hurts, never helps); what was wrong was treating a borderline, underpowered $n = 10$ result as a confirmed architectural property. We do not conclude independent DQN is immune to churn: replacing a trained agent's own decision-making with an untrained one plausibly degrades that agent's local performance regardless of whether it ever used cross-agent coordination, and the combined evidence across both samples now points that way.

C. Demand Spike: A Different, Cleaner Story

Fig. 5(c) shows no threshold at all on block precision, which stays flat and near-ceiling through the highest tested multiplier for every arm. The normalised-reward signal (not shown) that spike *improves* per-request reward is real and reproduces across both seed samples for GAT-CTDE, independent DQN, and the federated arm ($p \leq 0.014$ in both), but we do not read it as better decision-making: it traces to a specific, understood property of the existing reward shape (a fixed per-step violation penalty diluted by a larger denominator once request volume rises, still present after the primary volume normalisation above, which only corrects the dominant, much larger effect from the reward's accept-scaling service term) – block precision, a volume-invariant ratio, is the metric that actually answers whether spike corrupts decision quality, and it says no. Single-agent DQN's version of this same reward-based signal is significant in the original sample ($p = 0.0020$ at every severity) but not in the replication ($p = 0.084$) – direction consistent, significance sample-dependent, reported as such rather than as an unconditional finding.

D. Independent-Seed Replication

Every dropout result above, and GAT-CTDE/the federated arm's churn costs, were checked against the independent

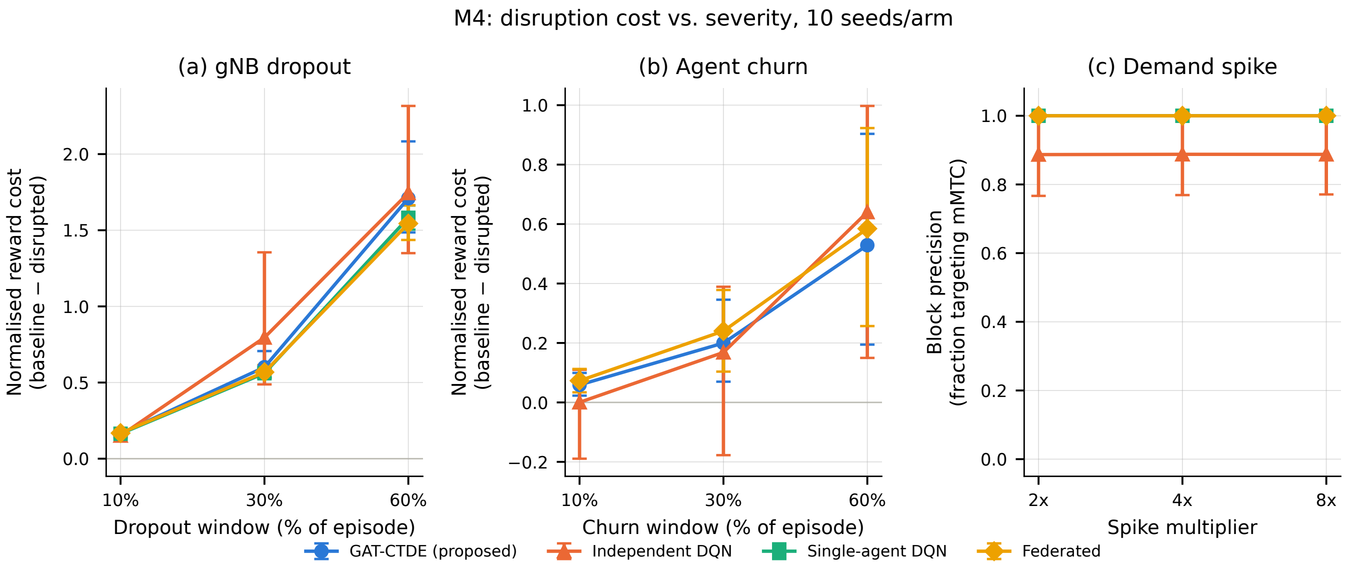

Fig. 5. M4 disruption campaign, 10 seeds/arm. (a)-(b) Normalised reward cost (baseline − disrupted) vs. severity for dropout and agent churn – every arm, every severity, shows a real cost that accelerates with severity rather than growing linearly. (c) Block precision vs. demand-spike multiplier: flat and near-ceiling for every arm, unlike (a)-(b) – spike does not threshold on the metric that matters.

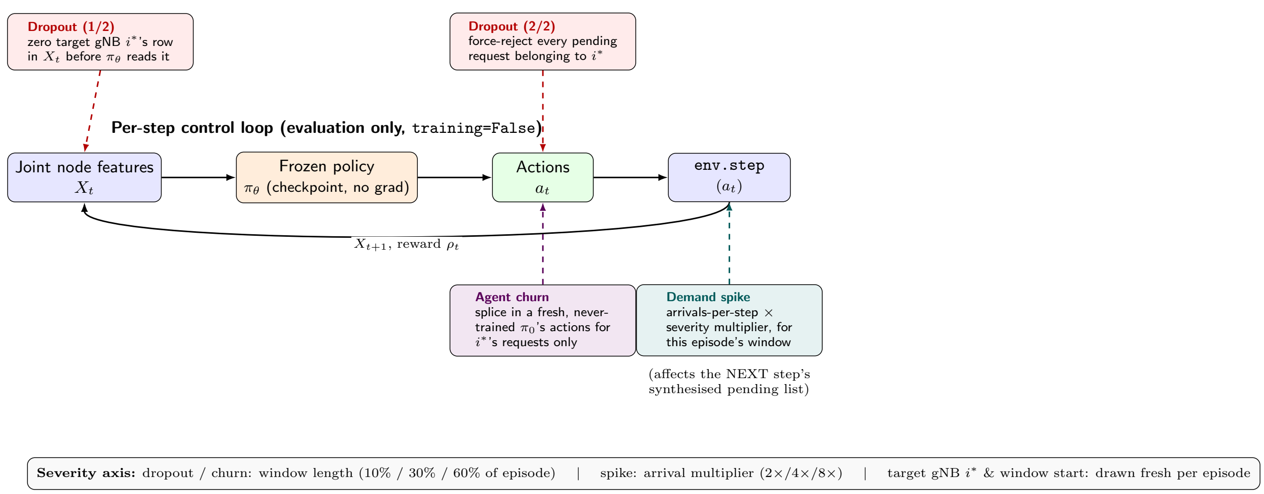

Fig. 6. Where each M4 disruption kind intercepts the frozen policy's per-step evaluation loop. Dropout acts twice (corrupting the target gNB's input row, then separately forcing that gNB's own requests to reject, so a policy cannot simply learn around the corrupted feature); agent churn substitutes a fresh, never-trained policy's actions for the target gNB only, leaving every other agent's decisions untouched; demand spike is entirely external to the policy, acting on the arrival process the environment synthesises for the following step. All three are evaluation-only perturbations – the policy itself is never updated.

1000–1009 sample and hold up in both direction and approximate magnitude – e.g. GAT-CTDE's dropout costs there are +0.170/+0.552/+1.449, close to the 900–909 sample's +0.165/+0.601/+1.710. The one finding that did not survive (independent DQN's supposed churn immunity) is reported only in its corrected form above, not as an initial claim we later walk back – this section already reflects what replication showed, exactly the discipline Section V-E argues for. One caveat the replication surfaced on the M3 privacy sweep (Section VIII) rather than here: block precision's degradation *exists* in both samples but the exact noise level it begins at does not (perfect through $\sigma = 1.0$ in one 10-seed sample, already degrading at $\sigma = 0.5$ in the other) – a threshold is real; its precise location is not a property we would claim generalises beyond the specific ten seeds it was measured on.

Fig. 7 puts every headline paired comparison in this paper next to its independent-sample counterpart on one axis, recomputed directly from each sample's own raw evaluation logs rather than retyped from either sample's summary numbers. Panel (a)'s three architecture/federation comparisons and panel (b)'s dropout/churn costs for GAT-CTDE and the federated arm all keep the same sign and a broadly overlapping confidence interval across samples – visually, the committed and fresh estimates sit close together on every row except one. The exception is independent DQN's churn cost (panel (b), bottom row, shaded): both samples agree the cost is positive (churn never helps), but the committed sample's interval reaches down toward zero while the fresh sample's sits clearly higher and to the right of every other churn estimate in the panel – the same reversal Section IX-D reports in numbers, here shown as the one row that visibly does not track its own sample-to-sample agreement the other six rows show.

E. A Formal Cross-Architecture Test

Every disruption cost so far compares one arm's disrupted performance against its own undisrupted baseline – not against another arm's cost directly. We close that gap with a paired Wilcoxon test on (GAT-CTDE's cost − independent DQN's cost) across the same 10 seeds, for every (kind, severity) cell, computed via the same normalised metric this section already uses (reused, not re-derived, from `m4_correctness_metrics.py`'s own per-seed function). Only one cell reaches significance: dropout at the mildest severity (10% window), where GAT-CTDE's cost is significantly *higher* than independent DQN's ($+0.0135$, 95% CI $[0.0053, 0.0236]$, Wilcoxon $p = 0.0059$, 9 losses / 0 ties / 1 win). Every other cell – dropout at 30%/60% and churn at all three severities – shows no significant difference ($p \geq 0.23$). The one significant result runs counter to a naive expectation that coordination should help more under disruption: at the mildest severity, GAT-CTDE's shared, attention-read joint state means even a brief corruption to one gNB's row is visible to every agent's decision, while independent DQN reads only its own row and is affected only when it is itself the dropped target – a plausible mechanism, not one this bounded test isolates directly. We report the result as it is: real at one severity, absent everywhere else tested, not a general claim that coordination costs more under disruption.

X. Results: Cluster-Size Scaling

A. Setup

We extend the cluster from the $N = 3$, fully-connected topology used throughout Sections VII–IX to $N \in \{7, 19\}$, the standard first- and second-ring cellular frequency-reuse cluster sizes, under three adjacency structures: fully-connected, a ring (N-cycle, degenerate to fully-connected at $N = 3$), and a hex-grid (concentric rings of cells around one centre, defined only for $N \in \{7, 19\}$). GAT-CTDE is the only arm whose Q-head consumes this adjacency; independent DQN and single-agent DQN never read a graph structure, and serve as a topology-invariance check on the two metrics below. We scale synthetic arrivals per step with N to hold offered load per gNB roughly constant; an early pass scaled arrivals but left the per-step pending-request cap at its $N = 3$ value, silently truncating $N = 19$'s intended contention to under two-thirds of its target rate. We caught this before trusting any $N = 19$ result, corrected the cap to scale with N, and independently re-verified the fix at the individual-decision level with a direct Q-value probe (below) before treating the corrected data as evidence.

We normalise reward per gNB before averaging, the identical fix Section IX already applies on the disruption-severity axis, since `action_mapping.py`'s accepted-request count sums across all gNBs in a step rather than per gNB, mechanically inflating raw reward with cluster size independent of decision quality. `sla_compliance_all_slices` is OR-aggregated across gNBs per step by construction (`reward.py`'s own documentation), so it trends toward 0 as N grows regardless of policy quality – a third instance, now on the cluster-size axis, of this paper's central evaluation-integrity finding; we report block precision and per-gNB- normalised reward as the primary metrics here for the same reason Sections VII–IX already do. The primary sample is 12 seeds (900–911) at $N = 19$, all three topologies, all three arms, judged the more promising thread to power up ahead of the broader topology-sparsity/reward-margin questions, which a smaller $N = 7$ pilot (3 seeds) already suggested were unlikely to show a strong effect. An independent-seed replication (1000–1002, disjoint) checks whether the $N = 19$ findings below generalise.

B. Reward Margin Does Not Grow With Cluster Size

An early reading of the $N = 7/N = 19$ pilot appeared to show GAT-CTDE's reward edge over single-agent DQN growing with N (raw reward diff +0.25 to +0.49 at $N = 7$, +0.67 at $N = 19$) – confounded by exactly the accept-volume-inflation mechanism above, and reversed once corrected (per-gNB-normalised diff +0.035–0.070 at $N = 7$, +0.035 at $N = 19$: flat, if anything slightly smaller). At the full 12-seed sample, the paired per-gNB reward-margin comparison at $N = 19$ is not significant for any topology: fully-connected -0.175 [95% CI -0.528, +0.076], $p = 0.7422$; ring -0.704 [-1.879, +0.042], $p = 0.7422$; hex -0.695 [-1.873, +0.053], $p = 0.9102$ (Wilcoxon, 6/4/2, 5/4/3, and 6/3/3 wins/ties/losses respectively) – the mean difference is, if anything, slightly negative for GAT-CTDE at this scale. This is mechanically explicable, not a red flag: single-agent DQN's always-accept collapse (next subsection) collects the reward's service term on every request with no rejection cost, while GAT-CTDE pays a real cost every time it correctly rejects mmTC – exactly the reward-optimal action per Section IV-A1's own calibration, just not one the aggregate reward signal clearly rewards at this N. This paper's evidence for GAT-CTDE's advantage at $N = 19$ lives in collapse resistance and block precision, not in the reward-margin metric – reporting a reward-margin advantage here would not be supported by what actually ran.

C. Collapse Resistance: A Real, Cluster-Size-Specific Effect

Fig. 8(a) reports collapse rate at $N = 19$. Single- agent DQN collapses totally: 12/12 primary-sample cells and 3/3 replication cells, every topology, zero exceptions across two disjoint seed samples (15/15 combined) – confirmed at the individual-decision level by a direct Q-value probe identical in method to Section V's original $N = 3$ diagnosis: $Q(\text{accept}) > Q(\text{reject})$ on 100% of 1,347 real pending-request decisions in a fresh greedy eval episode, all three slices, mean mmTC gap +29.6. GAT-CTDE collapses only partially: 11/36 (31%) in the primary sample, a first read the next subsection shows was itself only part of the picture. The same probe on GAT-CTDE's checkpoint shows unanimous accept-preference for urllc and

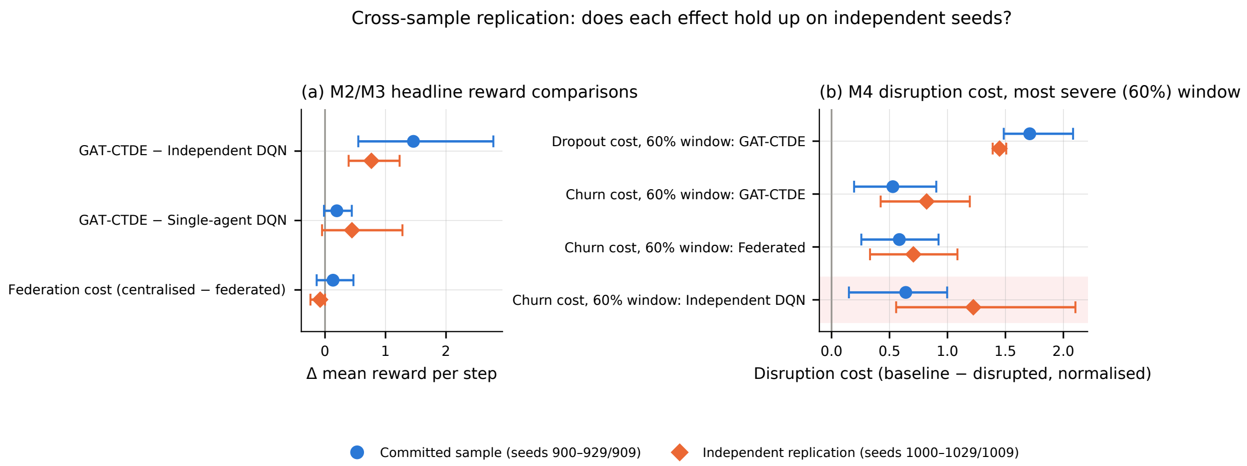

Fig. 7. Every headline paired comparison in this paper, committed sample (seeds 900–929/909) against the independent replication sample (seeds 1000–1029/1009), recomputed from raw logs for both. (a) M2/M3 reward comparisons. (b) M4 disruption cost at the most severe (60%) window. The bottom row of (b), independent DQN's churn cost, is the one comparison whose two samples visibly disagree on more than point-estimate noise – the finding retracted in Section IX-D.

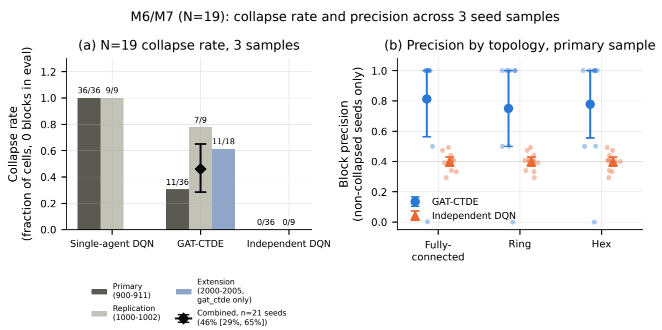

Fig. 8. $N = 19$ cluster, per-seed evidence across three independent seed samples. (a) Collapse rate (fraction of eval runs with zero blocks), pooled across all three topologies – single-agent DQN collapses totally in all three samples; GAT-CTDE's rate varies substantially sample-to-sample (primary 31%, replication 78%, a third sample targeted specifically at this question 61%), converging to a combined, seed-level-bootstrapped estimate of 46% [29%, 65%] across all 21 independent seeds (diamond marker, error bar). (b) Block precision by topology, primary sample, non-collapsed seeds only (single-agent DQN has none to show): GAT-CTDE's non-collapsed seeds are mostly near-ceiling but with a low-precision seed at every topology, while independent DQN sits in a narrow, mediocre band, identical across topologies as expected for an arm that never consumes adjacency.

eMBB and unanimous *reject*-preference for mmTC (mean gap -16.2) – the differentiated-shedding signature Section IV-A1 defines, confirmed at $N = 19$ and, if anything, more confident than the same probe's $N = 3$ result. Independent DQN never fully collapses at $N = 19$ (0/36) but never cleanly differentiates either: Fig. 8(b) shows its block precision in a narrow, mediocre 0.29–0.49 band, identical across all three topologies, as expected for an arm that never consumes an adjacency matrix. GAT-CTDE's own non-collapsed seeds show a topology-dependent secondary failure mode Fig. 8(b) makes visible directly: most hold near-perfect precision, but at least one seed per topology drops to 0.0–0.5, and one specific seed shows perfect precision (1.000) at fully-connected and near-zero (0.000) at ring/hex on the identical seed, environment stream, and initial weights – topology sparsity does affect GAT-CTDE's reliability, even though it does not affect its overall collapse rate.

D. Independent-Seed Replication and a Third Sample: Resolving GAT-CTDE's Collapse Rate

The replication sample (1000–1002) confirms single-agent DQN's total collapse exactly (3/3, every topology) and independent DQN's mediocre-but-never-collapsed profile (precisions 0.32–0.34, within the primary sample's 0.29–0.49 range, still topology-invariant), but GAT- CTDE's own collapse rate does not clearly replicate: 7 of 9 (arm, topology) cells (78%) collapse in the replication sample, notably higher than the primary sample's 31%. Both samples agree on the qualitative ordering – single-agent DQN always collapses, GAT-CTDE sometimes does, independent DQN never fully collapses but never cleanly differentiates either – but at $n = 3$ per topology, a real population rate of 78% and the primary sample's 31% cannot both be precise readings of the same fixed quantity from samples this small, so rather than report the range as an open question we drew a third, independent sample (2000–2005, 6 seeds) targeted specifically at this one question – not the broader topology-sparsity or reward-margin questions the previous subsection already answered from the first two samples.

The third sample collapses 11/18 (61%), between the first two. Pooling all three samples gives 29/63 cells collapsed (46.0%), but because collapse status for a given seed correlates across its three topologies (confirmed directly: several of the 21 seeds collapse at all three topologies or none, not a uniform scatter), we bootstrap the 95% confidence interval at the seed level – each of the 21 independent seeds contributing its own collapse fraction across its three topologies – rather than over the 63 pooled cells, which would understate the true uncertainty. This gives 46.0% [29%, 65%] (Fig. 8(a), diamond marker). Neither of the two smaller samples was wrong: both were honest readings of a genuinely wide distribution that happened to land on opposite sides of it. GAT- CTDE's collapse rate at $N = 19$ is best understood as roughly a coin flip with substantial seed-to-seed variability – not a tightly located rate the way single-agent DQN's 100% or independent DQN's never-collapses profile are – and three independent samples, not one, is what it took to state that with confidence.

XI. Results: Live Single-gNB Anchor

A. Setup

Prior live-testbed recalibration work on this rig (motivating the offline stress environment of Section III, though reporting no results of its own in this paper) already found that the offline environment cannot predict live *ranking*: Spearman $\rho = 0.097$ ($p = 0.855$) between offline and live compliance across six single-gNB checkpoints, unmoved by a 162-configuration recalibration grid search, with live compliance itself compressed near-ceiling for most of those checkpoints. We do not repeat that attempt. Instead we ask a narrower, bounded question this rig can actually answer: does the offline-trained *decision logic* – not its offline-reported outcome numbers – survive contact with real traffic at all. Only the admission-control question is testable here: this rig has one physical gNB, so GAT-CTDE's multi-gNB coordination benefit and the federated/DP results are architecturally untestable on it. We trained a fresh single-agent-DQN checkpoint against a single-gNB config dimension-matched to the live evaluation path (a checkpoint trained on this paper's 3-gNB configuration does not load into a single-gNB policy at all – confirmed directly, a real shape mismatch, not assumed) and ran it through the existing live-evaluation xApp against a real OAI gNB, 3 UEs, and real per-slice UDP traffic, twice: once from a backlog state left over from prior testing, once from a verified-clean baseline.

B. Decision Logic Transfers; the Margin Metric Does Not

Both live runs blocked only mmTC (46/32 requests in the first run, 45/32 in the second – zero urllc or eMBB blocks in either), exactly matching this same checkpoint's offline block precision of 1.000. The offline-trained differentiated-shedding decision transfers to live traffic essentially unchanged, replicated across two independent runs.

Both runs' SLA-margin readings, by contrast, diverged from anything the offline environment produces: eMBB's per-slice margin fell from $+1.0$ at the first live step to -122.4 within seconds and to $-1{,}002{,}377.5$ by the end of a 10-minute run – and this exact value recurred, to the decimal, in the independent second run. We first read this as contamination from an earlier, unrelated live test and nearly discarded it; the second run's identical result from a confirmed-clean starting state ruled that out; the eventual cause was a concrete, code-level unit mismatch, confirmed rather than assumed. `reward.py`'s SLA-margin computation is deliberately unclamped and divides a per-slice backlog reading by $L_{\max}$ (`kpm_adapter.py`). Offline, that backlog reading is a synthetic proxy, $\mathcal{N}(5.0, 2.0)$ (`replay_kpm_source.py`), calibrated to sit near $L_{\max} = 10$'s own scale. Live, the identically-named field is wired to the real MAC scheduler's RLC transmit-buffer byte count (`sched_ctrl->num_total_bytes`, confirmed directly against the framework's own source comment, not the offline model's synthetic distribution) – a physically real quantity that reaches into the millions of bytes within minutes once a sustained stream outstrips its PRB ceiling, and is divided by the same $L_{\max} = 10$ calibrated for a value two to three orders of magnitude smaller. The policy's own *input* state escapes this: `kpm_adapter.py` separately clips the state-facing backlog term to $[0, 2]$ before it ever reaches the network, while the margin diagnostic is deliberately left unclipped so it can expose exactly this kind of gap – which is why the policy's decisions stayed sensible live even as the margin number that would nominally score them exploded.

C. What This Means, Checked Against Prior Live Data

$L_{\max}$ is disconnected from live's real byte scale regardless of what caused any one run's specific numbers – but it is not simply "wrong." Checking 28 seeds of an already-committed prior live campaign on the same rig (the arm this paper's own live-testbed recalibration work draws its live target from) rather than assuming this diagnosis was already complete: 26 of 28 land on an embb margin of exactly 0.700, matching that recalibration work's independently- derived live target (eMBB mean 0.714) almost exactly, and – checked directly against that data's own logged limitations, not assumed – this is genuine backlog, not the same fallback proxy that field falls back to when the live sensor reads zero. $L_{\max} = 10$ reproduces a sensible number under conditions where a policy's ceiling keeps pace with demand; it only explodes once a real, already-documented failure regime is entered. `CAMPAIGN_LOG.md`, predating this session entirely, already measured this same regime directly: pinning embb's ceiling below organic demand raised backlog from a baseline of 411.3 to 6,601,079 (max 10,023,785) – matching this section's own catastrophic reading almost exactly – and recovery required widening the ceiling to five times the configured cap before backlog measurably drained. Whether this specific checkpoint's live run entered that regime for the same structural reason (a ceiling insufficient for organic demand) that a prior checkpoint's run apparently avoided is not fully resolved by this bounded anchor; the mechanism is real and independently documented, the full attribution is not.

D. A Latency-Normalised Recalibration

Rather than pick an arbitrary rescaling constant, or leave the mismatch as diagnosed-but-unpatched, we reused an already-built, previously-unused-for-this-purpose piece of this project's own calibration code: a Little's-Law byte-to-latency conversion, normalised against each slice's own already-calibrated latency budget instead of an arbitrary byte threshold – a physically meaningful quantity (an estimated queueing delay against a real SLA budget), not another guess. Backlog bytes are back-derived from the already-logged margin via `reward.py`'s own formula (not re-measured); service throughput uses `CAMPAIGN_LOG.md`'s own empirically-measured range for this run's ceiling (embb at max_ratio $= 4$ empirically serves

15–22 real PRB), stated as an approximation rather than a per-step reconstruction, since raw PRB-serving was not logged by this run. Recalibrated, the catastrophic case ($\approx$ 10.02MB backlog) becomes -809 to $-1{,}187$ (36–53 real seconds of queueing delay against a 45ms budget) rather than $-1{,}002{,}377.5$ – three orders of magnitude smaller and now directly interpretable – and the healthy reference (the prior campaign's 0.700) becomes 0.9996, revealing the original scale understated how good healthy conditions were, not only how bad severe ones looked. This does not make live and offline margins numerically interchangeable, nor does it re-open the rank-prediction question our live-testbed recalibration work already closed – offline's own worst case, bounded by its backlog capacity parameter, still sits roughly two orders of magnitude below even the recalibrated live catastrophic value – but it brings both onto the same order of magnitude rather than leaving a six-order-of-magnitude gap with no physical interpretation on either side.

XII. Discussion

Three findings in this paper are, we think, more broadly useful than the specific architecture: first, that a class-imbalanced, shared Q-head can silently suppress a minority slice's learning signal in a multi-slice admission controller, and that per-slice heads are a simple, verifiable fix; second, that an SLA-compliance metric which scores decisions only by whether they eventually keep a slice within its margin – not by whether the decision itself was correct given the information available at the time – can actively reward a degenerate policy over a correct one whenever the correct action cannot recover an already-stressed slice's margin. Any admission-control evaluation in a stress regime where this asymmetry holds should report a decision-level correctness metric alongside, not instead of, an outcome-level compliance metric.

Third, that reproducing a result on the same seed and replicating it on an independent one answer different questions, and only the second caught a real error in this paper's own analysis. Every number in this paper was checked for exact, same-seed reproducibility (an eval-log append bug and an unseeded weight-initialisation path, Section V-E, both surfaced and were fixed this way, neither by inspection alone), which confirms the pipeline is deterministic but says nothing about whether a finding is real or a property of one particular sample of seeds. Only the independent-seed replication in Section IX caught the latter: a claimed architectural immunity to agent churn that read as "mechanistically sensible" on one 10-seed sample was directly contradicted by a second, disjoint one. Section X's cluster-size campaign caught a second instance of the same pattern: GAT-CTDE's own collapse rate at $N = 19$ read as $\sim$31% on the primary sample and 78% on the replication one – a gap a third, targeted sample resolved (46% [29%, 65%], Section X) rather than one we reported as permanently open, but only because we kept sampling until the disagreement was explained rather than accepting either single-sample estimate as final. We would not have caught either gap by re-running the same seeds again, however many times, and we think any paper reporting an architecture-specific claim from a single seed sample – ours included, elsewhere, before this check – should weigh how much of its evidence is reproduction versus replication.

We are explicit about what remains offline-only. Every result except Section XI's bounded single-gNB anchor uses the offline stress environment (3, 7, or 19 gNBs depending on the campaign); none of the multi-gNB, federated, or disruption results are live claims, and Section III explains why we do not believe an offline-trained ranking here would transfer live without separate verification – a limit Section XI itself demonstrates directly rather than only argues for. Federated learning across three gNBs under one operator's cluster is also, as configured, a synthetic privacy scenario – genuine cross-organisation data separation would need multiple operators unwilling to pool raw data, which nothing in our current setup represents. We lead with the operational coordination benefit of periodic, not per-step, synchronisation (a real constraint for disaggregated near-real-time RAN Intelligent Controllers (near-RT RICs) with limited backhaul, even under one operator) rather than with a privacy motivation we cannot substantiate at this scale.

XIII. Conclusion and Future Work

We set out to test whether a standard evaluation metric can be trusted in multi-agent, privacy-constrained, disruption-prone slice admission control, and found it cannot go unchecked: a standard SLA-compliance metric, an unguarded append-only log, and an unseeded weight-initialisation path together hid three distinct failures from us – a training collapse the compliance metric scored no worse than correct behaviour, a hidden eval-data contamination bug, and a same-seed-only reproducibility gap – none of which a standard evaluation pipeline without the correctness-aware metrics and the reproduction-replication discipline this paper introduces would have caught. We built and iteratively fixed GAT-CTDE, the multi-agent architecture that served as this evaluation's substrate, and on the correctness-aware reward metric it delivers a decisive, statistically significant improvement over an independent-learner ablation, though its edge over our existing single-agent baseline specifically is small and not decisive at this sample size. Extending the architecture to federated training with differential privacy, we found federation to add no measurable reward cost and privacy costly only past a real threshold in decision precision, not a smooth trade-off. Evaluating the same frozen policies under mid-episode gNB dropout, demand spikes, and agent churn, we confirmed the threshold prediction this paper's own DP-noise finding suggested: dropout and churn both produce a real, accelerating cost with severity rather than graceful degradation, while demand spikes do not threshold at all on the metric that matters, a genuinely different failure mode. Cross-checking every result against an independent seed sample, not just the same seeds reproduced again, caught one architectural claim (that independent DQN is immune to agent churn) that a single sample had made look confirmed but was not – retracted before it could appear as a finding, with the correction reported alongside it rather than silently. Extending the cluster to 19 gNBs under ring and hex-grid adjacency, single-agent DQN's collapse becomes total and GAT-CTDE's remains only partial, a real cluster-size-specific effect distinct from anything at $N = 3$; an initial two-sample disagreement on GAT-CTDE's exact collapse rate (31% vs. 78%) was not left open but resolved with a third, independently-drawn sample targeted specifically at that question, converging on 46% [29%, 65%] across 21 seeds. Testing separately whether FedProx earns a measurable benefit under the genuine per-gNB load heterogeneity this investigation uncovered, we found none at any tested strength (two orders of magnitude in the proximal-term weight) – traced, not left unexplained, to real per-round client drift being too small relative to the loss scale for the proximal term to matter at this architecture's round length, a verified mechanism rather than an unexplained null.

Section XI took the most direct remaining step for the one architecture a single-gNB rig can test: the offline-trained admission-control decision logic transferred live essentially unchanged, while the SLA-margin metric did not, for reasons traced to a concrete unit mismatch rather than left as an unexplained gap. Whether the multi-gNB, federated, and disruption rankings in Sections VII–X would similarly survive contact with a live testbed remains untested – this rig has one physical gNB, so that question needs a multi-gNB rig this project does not currently have, not just more live time on this one. Whether the disruption-cost gap between coordination-dependent and independent architectures is itself statistically real is answered, not open: a formal paired test (Section IX) finds it real at only one of six tested cells (dropout's mildest severity, where GAT-CTDE costs significantly more, not less), absent everywhere else – a narrower, more qualified finding than either "coordination always costs more under disruption" or "the two architectures are equivalent." One narrower question this paper's own results raise is still open: why the privacy threshold's exact location shifted between our two seed samples (Section IX) – whether that reflects genuine sensitivity to which ten seeds are drawn, or a factor this paper has not identified. The FedProx-under- heterogeneity question this paper's own N-scaling investigation opened up is answered, not open, but raises its own narrower follow-up: whether a substantially longer inter-aggregation round (this paper tested only the existing 50-local-step default) would let real client drift accumulate enough for a heterogeneity dividend to emerge, a different variable than the one tested here and flagged rather than assumed.

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TODO(MANOJ): Same acknowledgment text as paper_conf/main.tex, or updated for this paper – please fill in.

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